



INF1005 - Web Systems and Technologies

Group Project

Overview

Objective

The purpose of this project is to provide students with practical experience applying the technologies and concepts learned in this module. You will design and develop a fully functioning, responsive, accessible, data-driven web application.

Organization

Working in teams, you are to build the website for a fictitious company/organization of your choosing (do not use real entities). You are free to have fun and be creative in selecting the type of business and the products and/or services it provides. However, your web application must satisfy the requirements below.

General Requirements

- A main landing page that introduces the company/organization.
- A common menu that allows the user to navigate to the various sub-pages.
- Sub-pages covering the products and/or services that the company provides.
- Sub-page that provides more details about the company/organization (e.g., "About Us").
- Back-end functionality supporting the creation, storing and updating of data (e.g., user membership, product catalogue, shopping cart, blog posts, etc.).

Technical Requirements

- Responsive, device-independent, mobile friendly design utilizing HTML5, Bootstrap and CSS.
- Custom JavaScript must be used for implementing dynamic client-side functionality.
- PHP and MySQL should be utilized to implement all back-end functionality, including comprehensive database operations (CRUD).
- Proper form validation and sanitization should be performed.
- Website should be secure from cross-site scripting and SQL injection attacks.
- User passwords should be protected appropriately.
- The website must adhere to W3C Web Validation and WCAG standards.

Optional Features

Beyond the minimum requirements, your team is free to implement whatever additional features you wish. You are advised to check with your instructor if you are considering using technologies not covered in the module. You may also refer to the FAQ in xSiTe for commonly asked questions.

Presentation

Teams will be required to demonstrate their website via a recorded presentation, due at the end of the final week of the trimester (check the Dropbox for exact date/time). The focus of the presentation will be on *individual contributions* to the project.

Presentation Guidelines

The team must present their project together as a whole, using their completed, published website. Take note of the following guidelines:

- Each group will submit a *single, cohesive video of the entire group* presenting together in one event, just as you would do if we were having live presentations. Do NOT submit separate files or cobble together videos recorded at different times.
- Each person should present for 3-5 minutes, with a maximum of 30 minutes for the team.
- As the focus is on individual contribution, **it's very important to state your name when it's your turn to present** so that we can attribute the work correctly. You should then proceed to demonstrate *only the parts that you worked on*.
- Do not use any PowerPoint slides or materials other than your website. Exceptions may include database diagrams and source code if you feel it better illustrates your contribution; otherwise, keep it to just the functionality of the website.

How to Submit the Presentation

A special Dropbox will be provided for each group to submit their team's presentation video, following the steps below:

1. Open the Dropbox and click on **Record Video**. Do NOT select Add a File as this has some limitations (you can still choose to upload your own file in the next step).

Project Presentation

▼ [Hide Folder Information](#)

Instructions

This Dropbox is for teams to submit their recorded Project Presentations. Please refer to the Project Guide for instructions on how to conduct the presentation and submit your video.

Submit Assignment

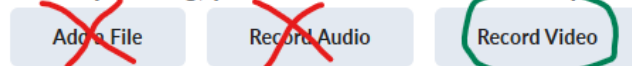
Allowed File Extensions

jpg, jpeg, png, gif, bmp, tif, tiff, swf, mpg, mpeg, rm, mp3, mp4, m4v, m4a, avi, wav, ram, asf, mov, ra

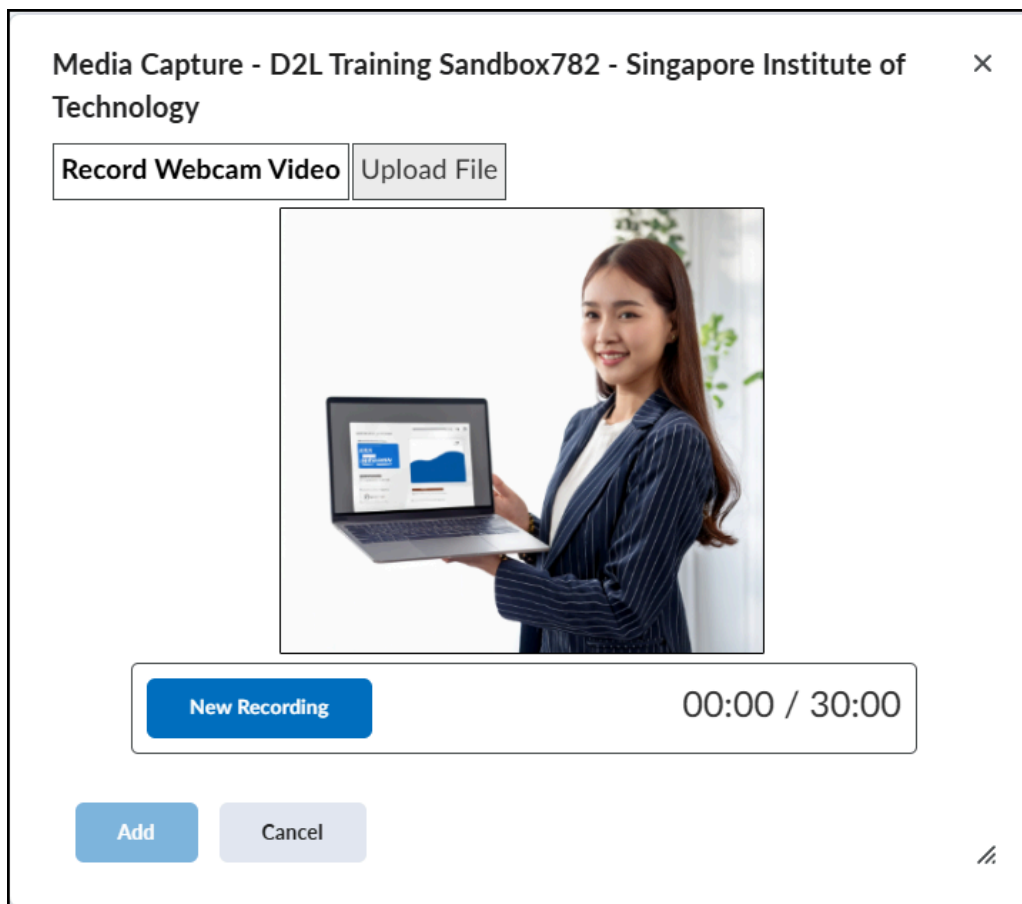
Files to submit *

(0) file(s) to submit

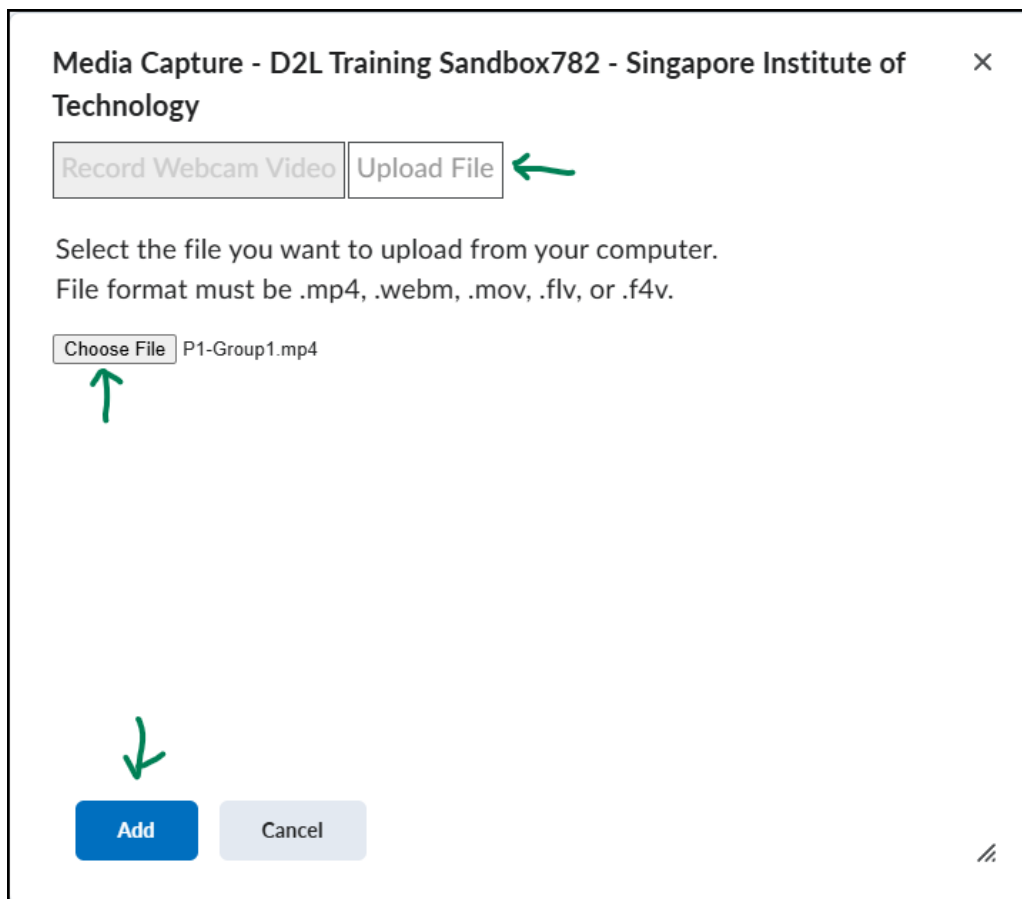
After uploading, you must click **Submit to complete the submission.**



2. On the next pop-up window, you can choose to either record your video directly to the Dropbox or you can upload a video file that you recorded separately.
 - a. To record your video directly, select **Record Webcam Video** and click on **New Recording** to begin recording using your computer's webcam, up to a maximum of 30 minutes. Be sure that both the presenter and the website can be seen within the frame.



- b. To upload a video file, select **Upload File** and then click the **Choose File** button.
Note: file format must be .mp4, .webm, .mov, .flv, or .f4v, and the size of the file cannot exceed 1 GB. You should name your file according to your section and group number, e.g. "P1-Group1.mp4".



3. After you record your video or upload your file, click the **Add** button and then in the next pop-up window, fill in the Title according to your section/group, e.g. "P1-Group1". Then click **Add** to add your video to the Dropbox.

Media Capture - D2L Training Sandbox782 - Singapore Institute of Technology

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Enter a video note description. A description is useful to users who are unable to view the video. You may also modify the title of the recording.

Title

P1-Group1

Description

Audio Language

English (United States) ▾

☐ Automatically generate captions from audio

↓

Add

Cancel

4. Finally, be sure to click **Submit** to complete your submission. Optionally, you can add any comments that you feel may be helpful to the evaluators. Note that this is a group submission, so any member may submit on behalf of the team.

Files to submit *

(1) file(s) to submit

After uploading, you must click Submit to complete the submission.

Add a File

Record Audio

Record Video

 [P1-Group1](#) 

Comments

Paragraph























Submit

Cancel

Report

Each group is also required to submit a report documenting their work. Requirements for the report are as follows:

- The report should contain a cover page with the names of the members in your group (full name and matriculation number), the module code, your group number, and the title of your project.
- The report should describe the design and implementation of the system. Include details of the tools and methodologies used by the team.
- Each team member should list in point form the parts of the website that they worked on.
- All credentials needed for logging into the website (as customer, admin, employee, etc.) for testing & assessment purposes should be listed in the report. Be sure that these credentials will work with *just username + password*, without need for 2FA or any external authentication (e.g. email or SMS).
- The Report must be saved in **PDF format**.

Submission of Website, Report, and URL

- Each group will be required to publish their website to a LAMP server running on Google Cloud or a similar publicly accessible cloud service. **This server must remain available for at least one week after the submission deadline**, to allow time to assess the website.
- In addition to hosting your site on the web server, you are also required to submit the following to the designated Dropbox on xSiTe:
 1. A .zip file containing your group's final integrated source code (follow the same procedure as with the Lab submissions). The code must be the same as what's running at your website URL.
 2. Your Report in .pdf format.
 3. The URL to your group's website.
- Only one submission per group is required and only the latest submission prior to the deadline will be considered. Do not submit individual versions of your work.
- Please take careful note of the submission deadline stated in the Dropbox - **late submissions will not be accepted**.

Peer Evaluation

Group members shall conduct peer reviews at the end of the project using the TEAMMATES application. Instructions for the peer eval will be released separately.

Policy Concerning Generative AI

Generative AI (gAI) tools, such as ChatGPT, GitHub Copilot, Gemini, etc., *may* be used, provided you adhere to the following conditions:

- gAI should only be used for generating specific functions or code fragments. For example, the following would be considered an acceptable prompt: "Write a function in PHP that takes a list of strings and sorts them in alphabetical order."
- gAI must *not* be used to create your entire project, whole web pages, or major portions of the work. For example, this prompt would NOT be acceptable: "Using PHP, HTML5, JavaScript and MySQL, create a standard e-commerce web application for selling cars."

- You are responsible for the correctness of all code submitted, including that generated by gAI, and you must understand and be able to explain all parts of your project.
- You are advised to check with your instructor before using gAI in your project.

Furthermore, if you utilize gAI tools when writing your Report, you *must* acknowledge this clearly and limit the usage of gAI content to no more than 10% of the entire paper. Refer to [How to Cite GAI](#) in the SIT Library guides for detailed policies that you should follow.

Assessment

The grade for the project will be comprised as follows:

	Group	Individual	Total
Completed Website	50%	10%	60%
Presentation	0%	20%	20%
Report	10%	0%	10%
Teamwork	0%	10%	10%
Total:	60%	40%	100%